

Rouge Construction: IPv6 Subnets

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Rouge Construction offers various construction services along with concrete services. In addition, I wanted to provide a network topology that supports the company's goal of expansion. To suit these services, I updated the topology from having a single subnet to an IPv6 network with four subnets to accommodate three different offices with room for expansion in the future. Two offices are dedicated to construction services, one for concrete services, and an additional subnet reserved for additional future services. In each subnet, there are no less than 3 devices including workstations and printers with room to accommodate many hosts. The network is connected via a dual-stack router so that devices and traffic that still use an IPv4 address can be accommodated to the IPv6 addresses. The router has an interface attached to both (configured) IPv4 and IPv6 addresses. This allows for hosts to communicate without changing IP addresses while having new hosts that use IPv6. Since the subnets are just for the 3 offices and a future office, the network prefix is 1234:2222:4567/62. Choosing the 62-bit for the prefix will open the opportunity for office expansion by providing 4 subnets with over a trillion hosts which can be applied to laptops, workstations, printers, and other machinery that may need access.

